

TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Skin-like covering for humanoid robots that adds tactile awareness

RoboSkin line from BeBop Sensors is a skin-like covering that adds tactile awareness to humanoid robots. It can



be fitted into the prosthetics, thus providing higher sensitivity and spatial resolution to the wearer. The sensor comes in a small package, which is less than a millimetre thick, making it the first available solution suitable for use in all robotic body parts, including fingers, limbs, head, torso, feet, etc. The Roboskin uses web based, real-time data visualisation and has multiple data visualisation modes that include live data charts and 3D renders.

BeBop Sensors

<https://bebopsensors.com>

Chip with near-sensor AI solution developed

GrAI VIP is an industry-first near-sensor AI solution with 16-bit floating-



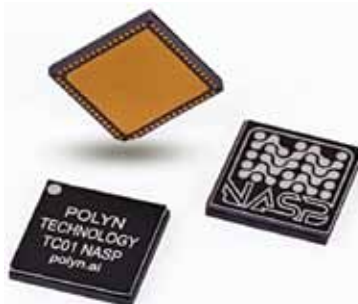
point capability. It has twenty times better power efficiency compared to alternatives available in the market. GrAI VIP provides a few millisecond inference latencies for networks like Resnet-50 and comes in a compact package of 8mm × 8mm with an integrated memory which enhances the privacy of your system. It achieves best in class performance and can deliver life-ready AI to industrial automation applications, such as pick and place robots, cobots, and warehouse robots. The chip has ready-to-use popular audio and vision networks for faster TTM.

GrAI Matter Labs

<https://www.graimatterlabs.ai>

Ultra-low-power neuromorphic analogue signal processor

Polyn Technology has announced their first neuromorphic analogue signal processor (NASP) chip. The ultra-low-power, high-performance chip can



replicate the pre-processing of the primary cortical area of the human brain. It is the first Tiny AI true analogue design to be used next to sensors. It can be used to demonstrate the proof of the technology's brain-mimicking architecture. The chip enables full data processing disaggregation between the sensor node and cloud. It finds usage in applications such as wearables,

Industry 4.0, Connected Health 4.0, etc.

Polyn Technology

<https://polyn.ai>

A processor that unifies functionalities of CPU, GPU, and TPU

Tachyum Prodigy from Tachyum is the world's first universal processor. Prodigy unifies the functionality of a CPU, GPU, and TPU in a single processor,



thus creating a homogeneous architecture and reducing performance degradation caused due to improper integration of different models. The universal processor is capable of delivering up to 6x performance for AI applications and 3x performance each for the best performing GPUs for HPC and highest-performing x86 processors for cloud workload available in the market. A universal processor can reduce power consumption and heat losses along with reducing the footprint

Tachyum

<https://www.tachyum.com>

World's first CXL memory expander controller

Montage Technology M88MX5891 is the world's first compute express link (CXL) memory eXpander controller (MXC). It is compatible with both the DDR4 and DDR5 JEDEC standards.